Chapter 2	Configuration Space
	2.1 DOF of a Rigid Body
	2.2 DOF of a Robot
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

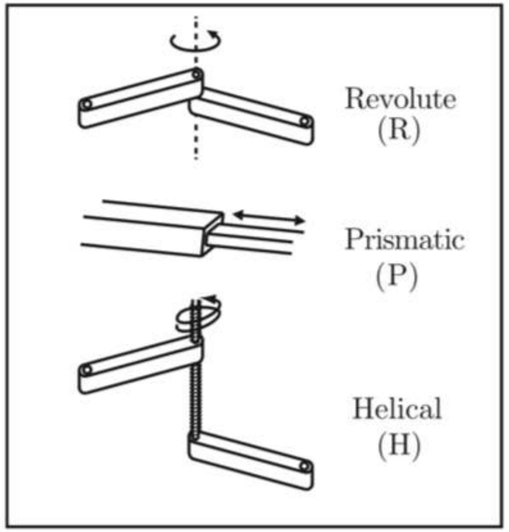
Important concepts, symbols, and equations

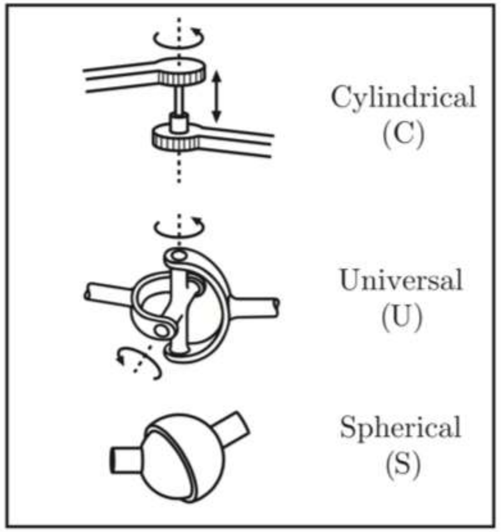
- **configuration**: a specification of the positions of all points of a mechanism
- **degrees of freedom** (dof): # of real #s required to describe a configuration
- **configuration space** (C-space): the dof-dimension space of all configurations
- dof of a planar body: $m = 3$; dof of a spatial body: $m = 6$

Important concepts, symbols, and equations (cont.)

- mechanism dof = Σ (body freedoms) – Σ (independent constraints from joints)

- joint types:



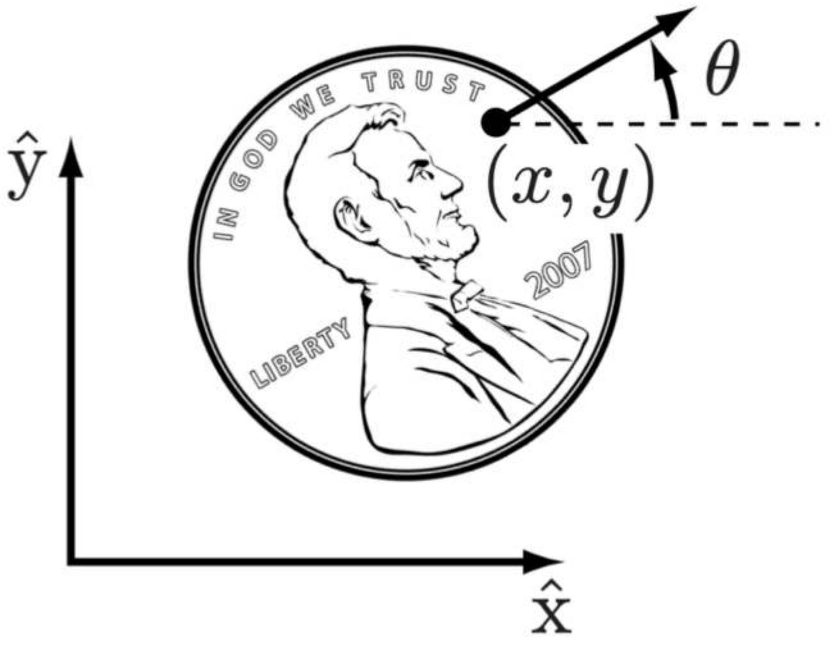


Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

- **Grübler's formula**: $\text{dof} = m(N - 1 - J) + \sum_{i=1}^{J} f_i$

→ # of freedoms provided by joints

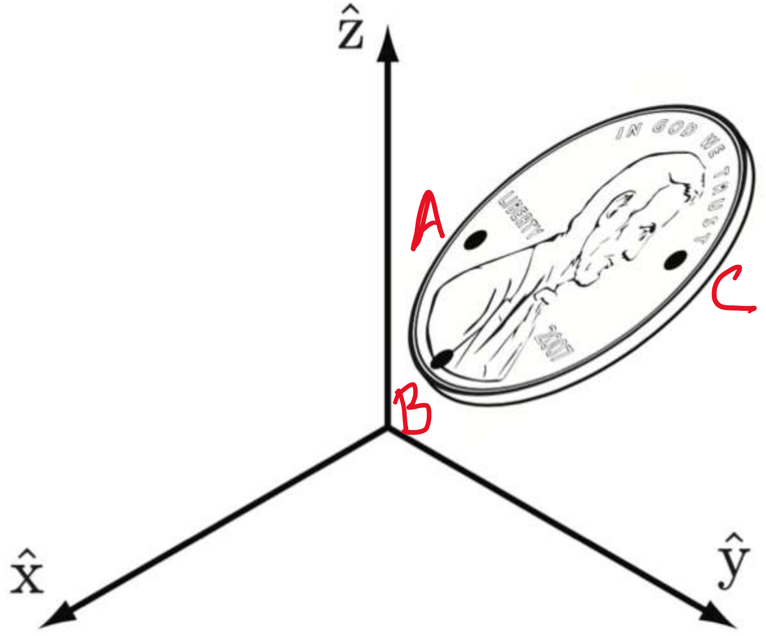
3 or 6   # bodies   # joints



coin on a plane

(x, y, θ) + $\underbrace{\text{heads/tails}}_{\text{not a real number}}$

$m = 3$

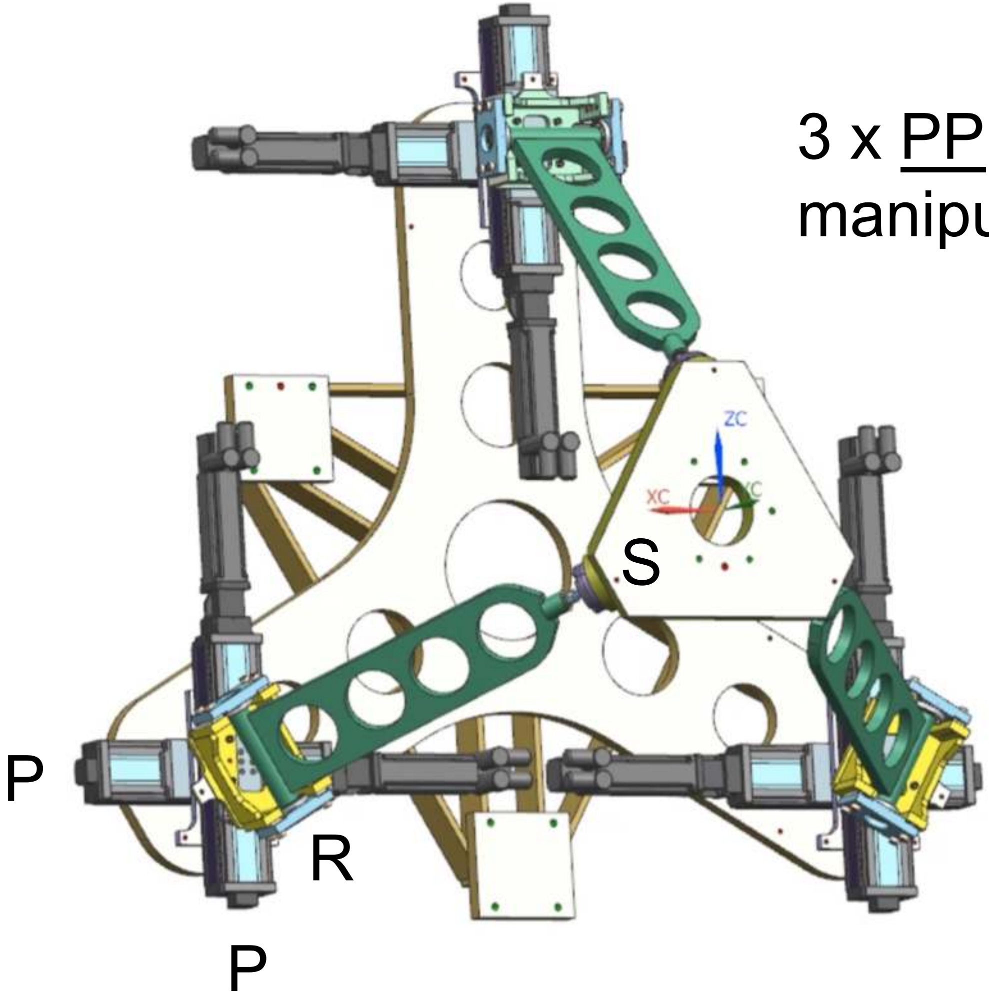


coin in space

A: 3 dof − 0 constraints = 3
B: 3 dof − 1 constraint = 2
C: 3 dof − 2 constraints = 1

$m = 6$

3 x PPRS parallel manipulator



KUKA Systems North America LLC
(patent pending)

$m = 6$ (spatial)

$N = 1 + 1 + 3 \times 3 = 11$

ground

platform

legs

2 prismatic carriages + link between R & S joints

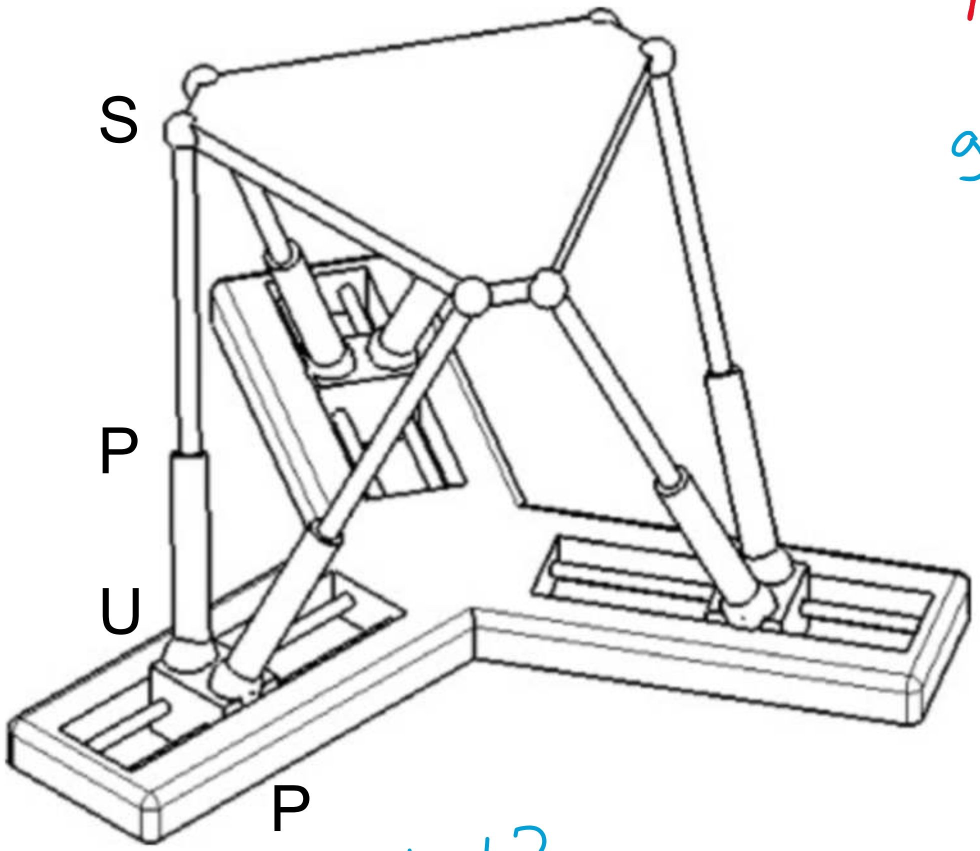
$J = 3 \times 4 = 12$

$\sum f_i = 3 \times (1+1+1+3) = 18$

$\text{dof} = 6(11 - 1 - 12) + 18$

$= -12 + 18 = 6$

$m = 6$ (spatial)



$$N = \underset{\text{ground}}{1} + \underset{\text{platform}}{1} + \underset{\text{sliders}}{3} + \underbrace{6 \times 2}_{\text{legs}} = 17$$

$$J = 3 + 6 \times 3 = 21$$

$$\sum f_i = 3 + 6 \times (2+1+3) = 39$$

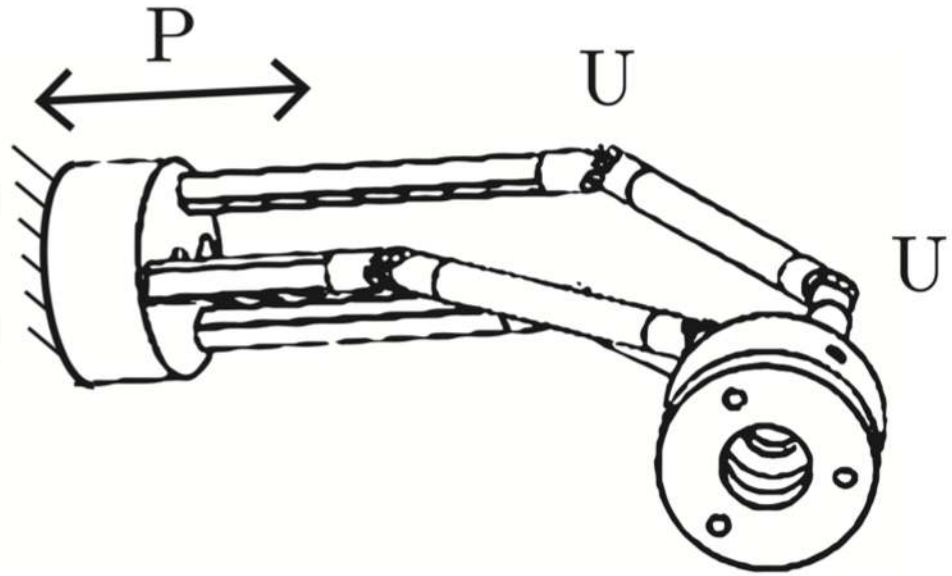
$$\text{dof} = 6(17-1-21) + 39$$

$$= -30 + 39 = \boxed{9}$$

underactuated?
redundantly actuated?

Which joints are likely actuated, and which are passive? How many dof does the platform have (not the mechanism)?

3 x <u>P</u>UU miniature surgical parallel manipulator
(National University of Singapore)

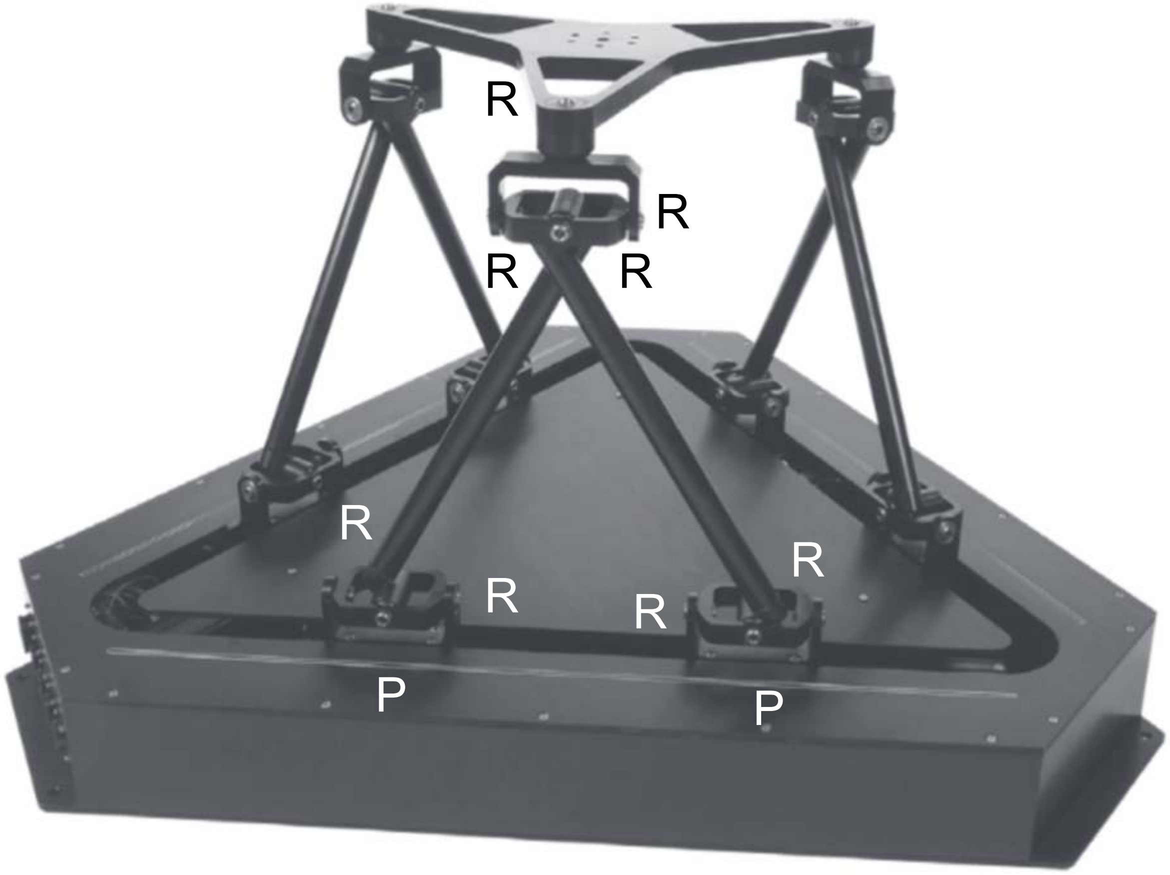


How many dof?

Discussion

underline of <u>P</u> indicates it is actuated. U joints are unactuated.

Quanser Hexapod



https://www.youtube.com/watch?v=AyVu4AE25DM

$m = 6$

$J = 3 \times 10 = 30$

$N = 1 + 1 + 6 \times 3 + 3 \times 2 = 26$

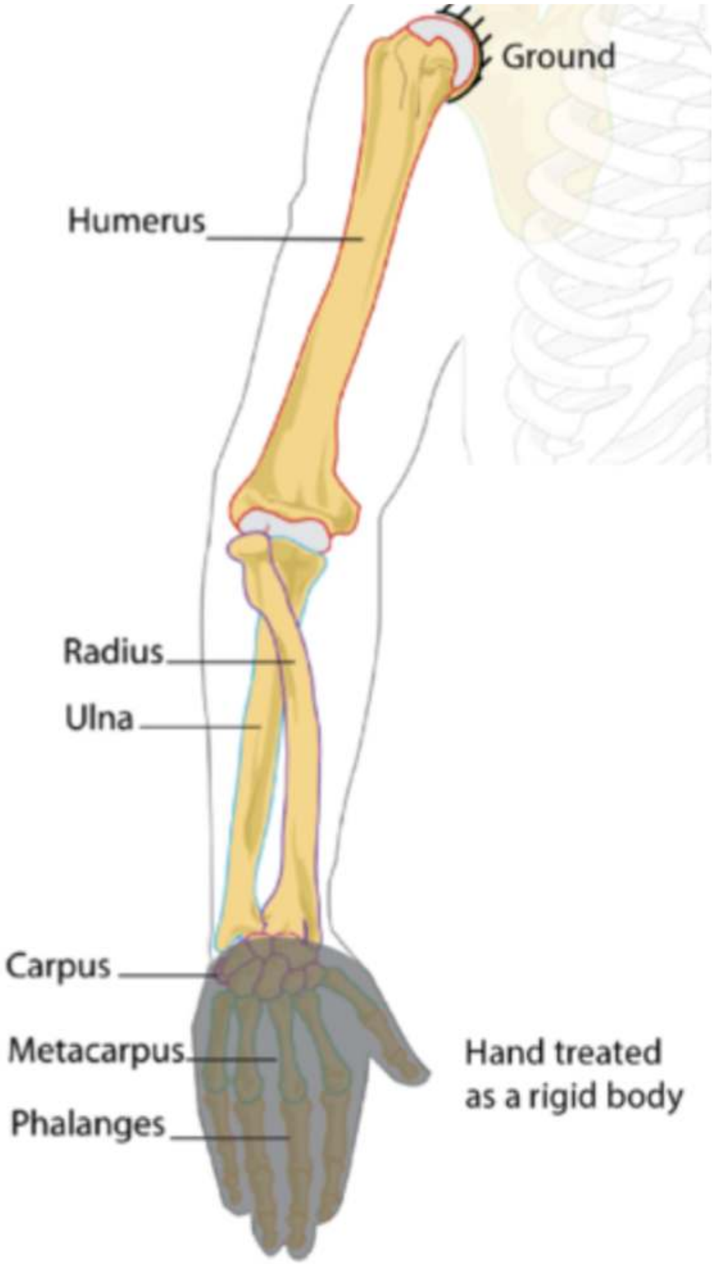
$\sum f_i = 30 \times 1 = 30$

$dof = 6(26 - 1 - 30) + 30$

$= 0$?

Clearly the platform has 6 dof.

Did we do something wrong?



How many dof does the human arm have?

Method 1: add dof of joints (shoulder, elbow, wrist)
Method 2: fully constrain hand's position

Discussion

How many total constraints are imposed by the joints?

Discussion